

Alisina Karimi

Software Engineer · Backend, Infrastructure & Developer Tooling | Turin, Italy | alisinakarimi.2003@gmail.com
Portfolio | [GitHub](#) | [LinkedIn](#) | Remote from Italy · Italian VAT contractor or EOR · no visa sponsorship required

Summary

Software engineer with a systems bias, already working remotely for an Australia-based commerce company from Italy. I own production services, AWS infrastructure, database migrations, CI/CD, observability, and integrations end to end. Strongest signals: PostgreSQL migration with **~60% infra-cost reduction**, **3×–10×** production query speedups, Terraform/Packer/ASG rollout work, and public projects with tests, containers, metrics, security hardening, and operator docs.

Experience

WineHub Commerce

Nov 2024 – Present

Software Engineer

Adelaide, Australia (Remote from Italy)

- Led MariaDB to **PostgreSQL** migration across schema design, query refactors, validation, and cutover; reduced infrastructure cost by **~60%** and followed with indexing/tuning that delivered **3×–10×** speedups on production workloads.
- Modernized the platform for horizontal scaling: externalized sessions to **Valkey**, moved async work to **AWS SQS**, replaced hardcoded IAM keys with instance-role credentials, and shipped ALB-aware health checks for multi-instance deployment.
- Authored **Terraform** for ASG, ALB target groups, scaling policies, lifecycle hooks, SQS, and ElastiCache; built GitHub Actions workflows for plan/apply and rolling ASG deployments.
- Built **Packer** AMIs on AL2023/ARM64 with EC2-tag-driven boot behavior, CloudWatch Agent, OpenTelemetry Collector, and Supervisor-managed worker pools.
- Designed **Go** services for analytics, billing, and webhook processing, while shipping core commerce work in **Symfony/PHP** and integrations with Sovos ShipCompliant, Avalara, Vinoshipper, and WineDirect.

Moneyviz srl

May 2024 – Nov 2024

Full Stack Developer

Turin, Italy

- Built financial analytics features in **Laravel** and **Vue.js** for a portfolio platform serving **1000+** user portfolios; improved ingestion/transformation paths to reduce dashboard latency.

RUN Polito APS-ETS

Sep 2022 – May 2024

Software Developer

Turin, Italy

- Built internal tooling in **Go** and JavaScript for **50+** members across 5 teams; automated recurring administrative workflows and refactored legacy services.

WEEEOpen, Politecnico di Torino

May 2022 – Aug 2022

Backend Developer

Turin, Italy

- Improved Django backend services and React integrations for an internal asset-management platform.

Selected Projects

Chatster | *Go, WebSockets, SQLite, React, Prometheus, Docker*

[github](#)

- Real-time chat stack with concurrent WebSocket hub, per-client write serialization, SQLite-backed replay, graceful shutdown, JSON logs, Prometheus metrics, rate limiting, Origin allowlist support, Docker Compose, and race-tested CI.

StreamHive | *Go, TCP/TLS, custom framing, content-addressed storage*

[github](#)

- Systems project with length-prefixed SHV1 frames, versioned handshake, context-aware listen/dial, TLS hooks, peer limits, BlobStore interface, liveness/readiness/metrics endpoints, pinned-action CI, coverage, and CycloneDX SBOM.

news-api | *TypeScript, Express, Redis, OpenAPI, OpenTelemetry, Kubernetes*

[github](#)

- Production-oriented API with Redis/in-memory caching, upstream timeouts, strict response validation, API keys, rate limiting, Pino logs, Prometheus metrics, OTLP tracing, OpenAPI, non-root Docker image, K8s probes, SBOM, and dependency-review CI.

Forge / md-files | *AI coding agents, prompt evals, safety hooks, Python tests*

[github](#)

- Installable Claude Code toolkit packaging specialist agents, progressive-disclosure skills, slash commands, safety hooks, docs, and an eval/test harness; useful signal for AI developer-tools and platform-engineering roles.

CARTOGRAPH | *TypeScript, ts-morph, architecture graphs, Git diffs, policy*

[github](#)

- Compiler-backed local analyzer producing deterministic architecture snapshots and evidence-linked semantic revision diffs, with bounded policy, report, runtime-reconciliation, and interchange contracts.

Open Source

Merged upstream pull requests in **pgx** (Go/PostgreSQL) and **Valkey** (C/distributed systems); active contributions to golang-migrate and Zed.

Skills

Programming: Go, PHP, Python, TypeScript/JavaScript, SQL

Backend: Symfony, Laravel, Express, Django, FastAPI, Gin, REST/OpenAPI, WebSockets

Cloud / Platform: AWS EC2/ASG/ALB/SQS/ElastiCache/Lambda, Terraform, Packer, Docker, Kubernetes, GitHub Actions, Linux

Data / Observability: PostgreSQL, MySQL/MariaDB, SQLite, MongoDB, Redis/Valkey, Prometheus, Grafana, OpenTelemetry, Pino/slog

AI / Security: AI developer tooling, agent/tool design, eval harnesses, detection engineering, DFIR, reverse engineering, scikit-learn, XGBoost, TF-IDF, embeddings; author of SignalForge, a cross-LLM security toolkit

Languages

English and Persian (bilingual) · Italian and French (fluent) · German (professional)

Education

Politecnico di Torino

Turin, Italy

B.Sc. Computer Engineering, 2021 – 2025